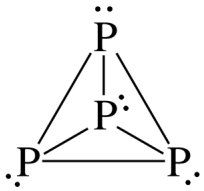
Question 3: Long Answer — 10 points

A For the correct diagram: **Point 01**



B (i) For a correct explanation: **Point 02**

Because gas particles are more dispersed (have more microstates) than solids, the entropy decreases as the reactants (which include a gas) convert to the solid product.

(ii) For the correct answer and a valid justification: **Point 03**

Yes. Given that $\Delta G^\circ_{rxn} = \Delta H^\circ_{rxn} - T\Delta S^\circ_{rxn}$*, the reaction must have* $\Delta G^\circ_{rxn} < 0$ *to be favorable. Because the reaction is exothermic,* $\Delta H^\circ_{rxn} < 0$ *and enthalpy contributes to favorability.* $\Delta S^\circ_{rxn} < 0$*, so entropy does not contribute to favorability.*

C (i) For the correct calculated value reported with the correct number of significant figures: **Point 04**

$q = mc\Delta T = (100.1\text{ g})(4.18\text{ J/(g}\cdot\text{°C)})(22.38\text{°C} - 22.00\text{°C})$

$q = 160\text{ J} = 0.16\text{ kJ}$

(ii) For the correct calculated value, consistent with part C (i): **Point 05**

$q_{rxn} = -q_{surr} = -0.16\text{ kJ}$

$$\Delta H^\circ_{rxn} = \frac{-0.16\text{ kJ}}{0.100\text{ g P}_4\text{O}_{10}} \times \frac{283.9\text{ g P}_4\text{O}_{10}}{1\text{ mol P}_4\text{O}_{10}} \times \frac{1\text{ mol P}_4\text{O}_{10}}{1\text{ mol}_{rxn}} = -450\text{ kJ/mol}_{rxn}$$

For the correct sign: **Point 06**

-450 kJ/mol_{rxn}

D For the correct answer and a valid justification: **Point 07**

Less than. If less P_4O_{10} *is present, less thermal energy will be transferred to the water during the reaction, causing the temperature increase to be less than it was with* 0.100 g *of* P_4O_{10}.

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E	For the correct calculated value: *Using Hess's law:* $\Delta H^\circ_{f,\,PCl_5(g)} = \frac{1}{4}\Delta H^\circ_1 + \Delta H^\circ_2$ $\Delta H^\circ_{f,\,PCl_5(g)} = \frac{1}{4}(-1148) + (-88) = -375$ kJ/mol	**Point 08**
F	(i) For the correct calculated value: $K_p = \frac{P_{PCl_5}}{(P_{PCl_3})(P_{Cl_2})} = \frac{4.00}{(2.00)(6.00)} = \frac{1}{3} = 0.333$	**Point 09**
	(ii) For the correct answer and a valid justification: *Decrease. The negative value of ΔH°_2 indicates that the reaction is exothermic. Because exothermic reactions favor reactant formation at higher temperature, the value of K_p decreases.*	**Point 10**

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